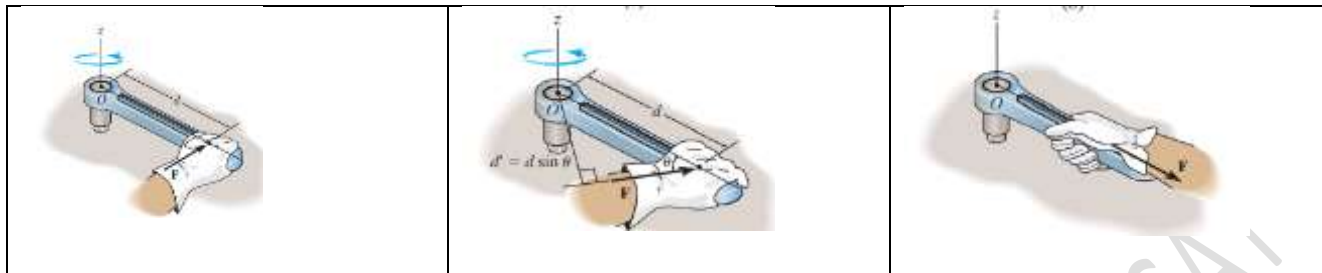


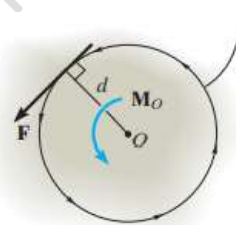
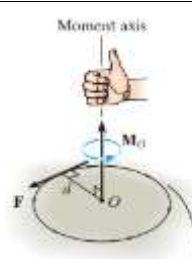
Moment of a force

The tendency of a force to rotate a body about an axis is known as the moment of the force about that axis. Consider the case of a force F applied to the wrench to turn a nut about the z – axis.



The magnitude of the moment is directly proportional to the magnitude of F and the perpendicular distance or moment arm d . If the force F is applied at an angle $\theta \neq 90^\circ$, then it will be more difficult to turn the bolt since the moment arm $d' = d \sin \theta$ will be smaller than d . If F is applied along the wrench, its moment arm will be zero since the line of action of the force intersects with the z -axis. Hence the moment of F about z -axis is zero.

The direction of the moment is given by the right-hand rule. The moment is considered positive if it is acting counter-clock wise.



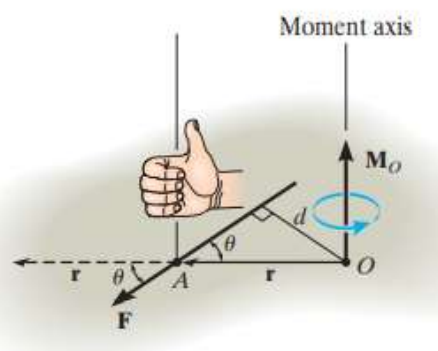
Moment of the force F about the axis passing through O and perpendicular to the plane can also be expressed

as the cross product,
$$\vec{M}_O = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

$\vec{r}$ is the position vector of any point A on the line of action of the force.

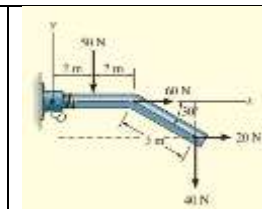
$$\vec{r} = r_x \hat{i} + r_y \hat{j} + r_z \hat{k}$$

$$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}$$



Example 1

Determine the net moment of the forces about the z -axis passing through O .



Moment of 50 N force about z -axis through O , $M_1 = -50 \times 2 = -100 \text{ Nm}$

Moment of 60 N force about z -axis through O , $M_2 = 0$

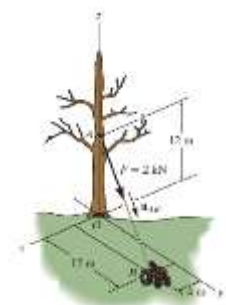
Moment of 20 N force about z -axis through O , $M_3 = +20 \times 3 \sin 30 = +30 \text{ Nm}$

Moment of 20 N force about z -axis through O , $M_4 = -40 \times (4 + 3 \cos 30) = -264 \text{ Nm}$

The net moment of the forces about the z -axis passing through O , $\Sigma M = -334 \text{ Nm}$

Example 2

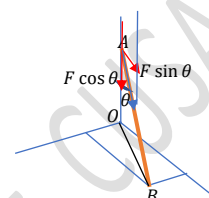
Determine the moment of the force about O.



$$OB = \sqrt{12^2 + 4^2} = 12.65 \text{ m}$$

$$\tan \theta = \frac{OB}{OA} = \frac{12.65}{12} \gg \gg \theta = 46.5^\circ$$

$$M_o = F \sin \theta \times 12 = 2 \times \sin 46.5 \times 12 = 17.4 \text{ Nm}$$



$$\vec{r}_A = 12 \hat{k} \ll \gg \vec{r}_B = 4 \hat{i} + 12 \hat{j} \ll \gg \vec{r}_{AB} = \vec{r}_B - \vec{r}_A = 4 \hat{i} + 12 \hat{j} - 12 \hat{k}$$

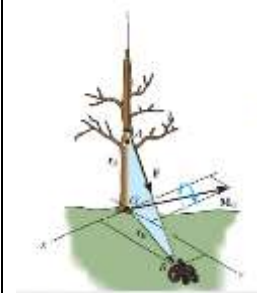
The unit vector along AB, $\hat{u} = \frac{\vec{r}_{AB}}{|\vec{r}_{AB}|} = \frac{4 \hat{i} + 12 \hat{j} - 12 \hat{k}}{\sqrt{4^2 + 12^2 + 12^2}} = 0.2294 \hat{i} + 0.688 \hat{j} - 0.688 \hat{k}$

The force vector, $\vec{F} = F \hat{u} = 2 \text{ kN} \{0.2294 \hat{i} + 0.688 \hat{j} - 0.688 \hat{k}\} = 0.4588 \hat{i} + 1.376 \hat{j} - 1.376 \hat{k} \text{ N}$

$$\vec{M}_o = \vec{r}_A \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_{Ax} & r_{Ay} & r_{Az} \\ F_x & F_y & F_z \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 12 \\ 0.4588 & 1.376 & -1.376 \end{vmatrix} = -16.5 \hat{i} + 5.51 \hat{j} \text{ kNm}$$

$$\vec{M}_o = \vec{r}_B \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_{Bx} & r_{By} & r_{Bz} \\ F_x & F_y & F_z \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 12 & 0 \\ 0.4588 & 1.376 & -1.376 \end{vmatrix} = -16.5 \hat{i} + 5.51 \hat{j} \text{ kNm}$$

Magnitude of the Moment, $|\vec{M}_o| = \sqrt{(-16.5)^2 + 5.51^2} = 17.4 \text{ kNm}$



1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson

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